Jacob Ke

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Professional Summary

Detail-oriented and analytical second-year student pursuing a Bachelor of Science in Mathematics & Statistics at the University of Toronto. Proficient in Python (Pandas, NumPy), R, SAS, and SQL, with experience building predictive models and analyzing large datasets. Strong foundation in statistical modeling, machine learning, and data-driven problem solving, complemented by hands-on experience in AI and data science projects.

Technical Skills

Programming & Databases: Python (Pandas, NumPy), R, Java (OOP), SQL (MySQL)
Data Analysis: Statistical Modeling, Hypothesis Testing, Linear & Logistic Regression, Feature Engineering
Machine Learning: Decision Trees, Random Forest, Logistic Regression, Cross-Validation, Model Evaluation
Tools: Git, PyCharm, IntelliJ, LaTeX, Microsoft Office
Soft Skills: Leadership, Team Collaboration, Problem Solving, Communication, Time Management

Internship Experience

Data Science Intern May 2024 – Aug 2024
IBM

- Designed and deployed a **production-oriented automated video generation pipeline** using Koze, integrating prompt engineering with Python.
- Optimized prompt structures and implemented dynamic content adaptation, resulting in a **50% increase in user engagement and usage**.
- Built and validated a **credit card fraud detection pipeline** using Logistic Regression, including end-to-end data preprocessing, feature selection, and feature scaling.
- Achieved **90% classification accuracy** on imbalanced datasets, with evaluation using precision-recall trade-offs to ensure robustness in real-world risk scenarios.

Project Manager, Facial Expression Accuracy Model Jan 2026 – Apr 2026
University of Toronto

- Performed dataset cleaning, duplicate removal, preprocessing, and leakage prevention on **450** labeled animal facial-expression images for CNN training.
- Developed and optimized transfer-learning CNN pipelines using fine-tuned **ResNet18**, **EfficientNet-B2**, multi-layer classification heads, **TrivialAugmentWide** augmentation, and **Focal Loss**, achieving **88.52%** validation accuracy and **83.33%** test accuracy.
- Through backbone replacement, classification-head redesign, and loss-function optimization, improved validation accuracy from **58.89%** to **88.52%**.

Application Designer May 2025 – Aug 2025
University of Toronto

- Co-designed and developed a cross-platform **Java** desktop application integrating adaptive flashcard algorithms, lexical reference systems, and lightweight gamification for foreign-language vocabulary learning.
- Implemented core functionalities including **user authentication**, password management modules, and activity-history tracking, improving system usability and persistent data management.

Academic Projects

Project Leader – Geography-Poverty Project Jan 2024

- Analyzed the relationship between food insecurity and geographic factors in Sub-Saharan Africa.
- Cleaned and processed datasets in R; developed classification and regression models to identify key predictors of hunger and Conducted comparative regional analysis using linear regression to uncover structural disparities.

Data Analysis Project Nov 2025

- Investigated the impact of flight characteristics and booking timing on ticket pricing using regression models with interaction terms.
- Identified key variable interactions influencing pricing across service classes.

Education

University of Toronto, St. George Campus
Honours Bachelor of Science – Mathematics & Its Applications Specialist (Statistics) Sep 2023 – Present
Mathematics & its Applications Specialist (ASSPE1758, ASSPE1890; ASSPE1580) Sep 2023 – Present
Relevant Coursework: Linear Algebra, Advanced Calculus, Probability , Machine Learning , Software Design , Advanced Programming.